EXPERIMENT 6

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Semester: 5
Subject Name: Computer Network **Subject Code: 24CSH-337**

Experiment 06

Implementation of VLAN and VLAN Trunking Protocols

Aim

To implement VLAN and VLAN Trunking protocols in Cisco Packet Tracer and understand how VLANs provide logical separation of networks and how trunk links carry traffic belonging to multiple VLANs between switches.

Objective

- To understand the concept of VLANs.
- To understand the difference between Layer 2 and Layer 3 switches.
- To configure VLANs on Cisco switches.
- To configure access ports and assign them to VLANs.
- To understand the concept of VLAN trunking.
- To configure a trunk link between two switches.
- To understand IEEE 802.1Q VLAN tagging.
- To verify communication between devices belonging to the same VLAN.
- To verify that devices in different VLANs cannot communicate without Layer 3 routing.

Software Requirements

- Cisco Packet Tracer

Hardware (Virtual)

- Cisco 2960 Switches
- PCs
- Copper Straight-Through Cables

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Theory

A Layer 2 switch primarily operates at the Data Link Layer and uses MAC addresses to forward Ethernet frames. A Layer 3 switch can perform both Layer 2 switching and Layer 3 routing and can therefore perform Inter-VLAN Routing.

VLAN stands for Virtual Local Area Network. It allows a single physical switch to be logically divided into multiple separate networks. For example, VLAN 10 can be assigned to one department and VLAN 20 to another department. Devices belonging to different VLANs are logically separated and normally require Layer 3 routing to communicate with each other.

VLANs help reduce broadcast traffic, provide logical separation between departments, improve network organization, and allow multiple logical networks to use the same physical switch infrastructure.

An access port normally belongs to one VLAN and is generally used to connect end devices such as PCs, printers, and servers.

A trunk port carries traffic belonging to multiple VLANs over a single physical link, and is generally used between network devices such as switches.

When VLAN traffic crosses a trunk, VLAN information is carried using VLAN tagging. The commonly used standard is IEEE 802.1Q, which allows the receiving switch to identify the VLAN to which a frame belongs.

Practical / Experiment Steps

1. Open Cisco Packet Tracer.
2. Place two Cisco 2960 switches and four PCs in the workspace.
3. Connect the devices using Copper Straight-Through cables:
   - PC1 → SW1 Fa0/1
   - PC2 → SW1 Fa0/2
   - PC3 → SW2 Fa0/1
   - PC4 → SW2 Fa0/2
4. Connect the two switches:
   - SW1 Gi0/1 → SW2 Gi0/1
5. Assign IP addresses to the PCs:

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PC1

- IP Address: 192.168.10.11
- Subnet Mask: 255.255.255.0

PC3

- IP Address: 192.168.10.12
- Subnet Mask: 255.255.255.0

PC2

- IP Address: 192.168.20.11
- Subnet Mask: 255.255.255.0

PC4

- IP Address: 192.168.20.12
- Subnet Mask: 255.255.255.0

6. Configure VLAN 10 and VLAN 20 on SW1.
7. Assign Fa0/1 of SW1 to VLAN 10.
8. Assign Fa0/2 of SW1 to VLAN 20.
9. Repeat the VLAN and access-port configuration on SW2.
10. Configure Gi0/1 on SW1 as a trunk port.
11. Configure Gi0/1 on SW2 as a trunk port.
12. Test communication between PC1 and PC3. Both devices belong to VLAN 10, so the ping should be successful.
13. Test communication between PC2 and PC4. Both devices belong to VLAN 20, so the ping should be successful.
14. Test communication between PC1 and PC4. Since they belong to different VLANs and no Layer 3 routing is configured, the ping should fail.

VLAN Configuration Commands

On each switch:

enable

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configure terminal

vlan 10

name STUDENTS

exit

vlan 20

name FACULTY

exit

Access Port Configuration

For Fa0/1:

interface fa0/1

switchport mode access

switchport access vlan 10

exit

For Fa0/2:

interface fa0/2

switchport mode access

switchport access vlan 20

exit

Trunk Configuration

On both switches:

interface gigabitEthernet 0/1

switchport mode trunk

exit

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These configurations correspond to the complete SW1 and SW2 configurations provided in the experiment.

Verification Commands

To view the configured VLANs:

show vlan brief

To verify the trunk configuration:

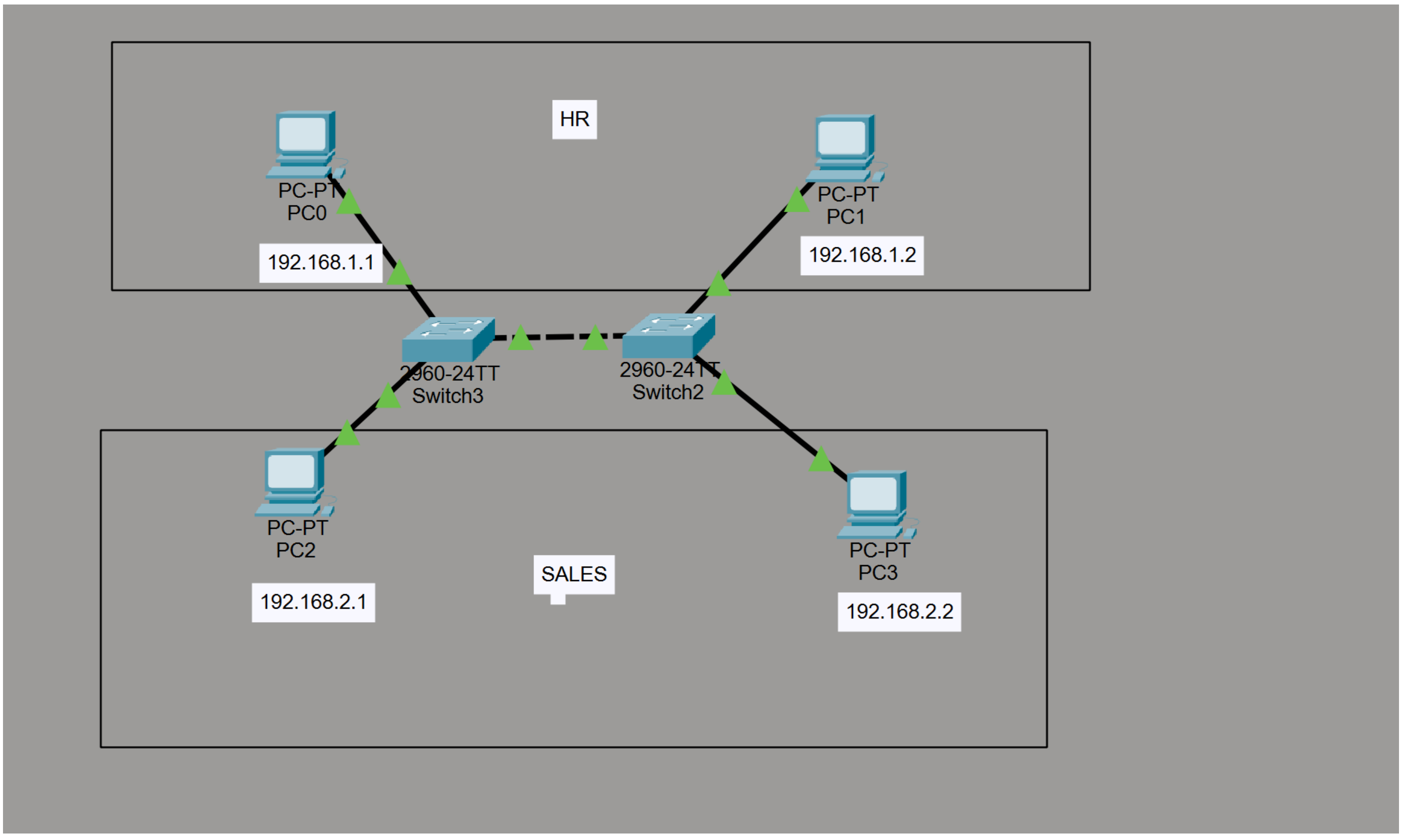
show interfaces trunk

To view the MAC address table:

show mac address-table

Output

Screenshot 1 – Complete network topology



Screenshot 2 – VLAN configuration on Switch 1

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```
Switch(config)#interface range fast
Switch(config)#interface FastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#sh vlan brief
                ^
% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#sh vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   HR                               active    Fa0/1, Fa0/2
20   SALES                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#conf t
```

Screenshot 3 – VLAN configuration on Switch 2

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```
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#sh vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   HR                               active    Fa0/1, Fa0/2
20   SALES                            active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Screenshot 4 – Trunk configuration

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```
% Invalid input detected at '^' marker.

Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchmode mode access
                        ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/1 VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/1 on VLAN0001. Inconsistent port type.


Switch(config)#conf t
                ^
% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#i
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
```

Screenshot 5 – VLAN and MAC address verification

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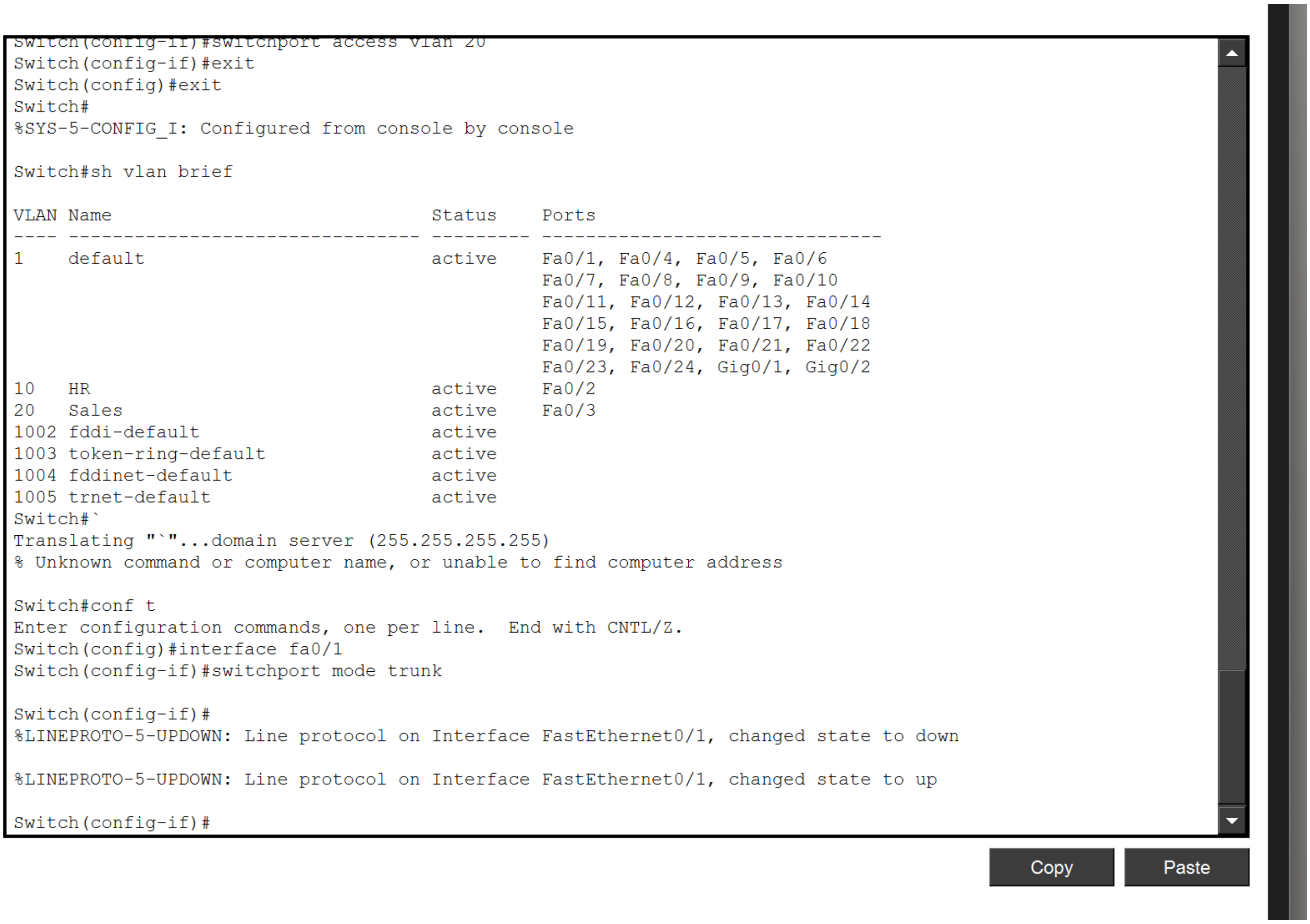
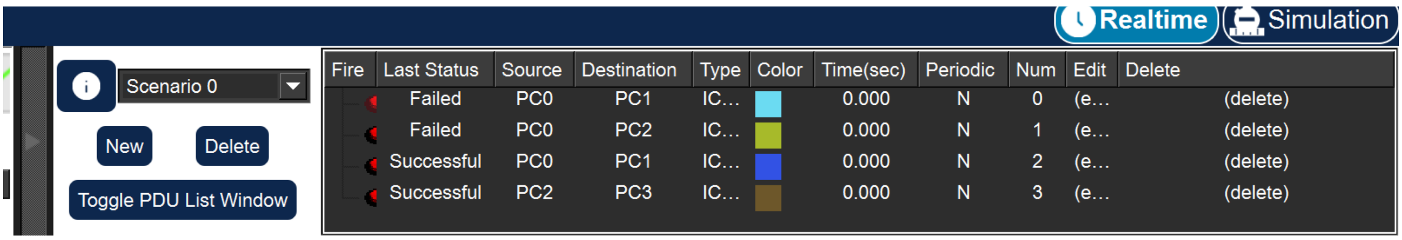


```
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#sh vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   HR                               active    Fa0/2
20   Sales                            active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#`
Translating "`"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#
```


Screenshot 6 – Successful Ping between devices in the same VLAN



Learning Outcome

After completing this experiment, I was able to:

- Understand the concept and purpose of VLANs.
- Understand the difference between Layer 2 and Layer 3 switches.
- Configure VLANs on Cisco switches.
- Configure access ports and assign them to specific VLANs.
- Understand the concept of VLAN trunking.

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- Configure a trunk link between two switches.
- Understand the purpose of IEEE 802.1Q VLAN tagging.
- Verify VLAN and trunk configurations using Cisco IOS commands.
- Verify communication between devices belonging to the same VLAN.
- Understand why devices in different VLANs require Layer 3 routing for communication.